

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Supplementary Summer Examination – 2023

Course: B. Tech. Branch : Mechanical Engineering Semester : Fifth

Subject Code & Name: BTMC503 Theory of Machines- II

Max Marks: 60

Date: 11/08/2023

Duration: 3.00 Hrs.

Instructions to the Students:

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

(Level/CO) Marks

Q. 1 Solve Any Two of the following.

- | | | |
|--|---------------------|-----------|
| A) Explain with neat sketch the phenomena of 'slip' and 'creep' in a belt drive. | Understand/
CO1 | 06 |
| B) Deduce an expression for the length of belt in an open belt drive. | Analyzing /
CO1 | 06 |
| C) A pulley is driven by a flat belt, the angle of lap being 120° . The belt is 100 mm wide by 6 mm thick and density 1000 kg/m^3 . If the coefficient of friction is 0.3 and the maximum stress in the belt is not to exceed 2 MPa, find the greatest power which the belt can transmit and the corresponding speed of the belt. | Evaluating /
CO1 | 06 |

Q.2 Solve Any Two of the following.

- | | | |
|--|---------------------|-----------|
| A) Define the terms (i) Module (ii) Pressure angle (iii) Addendum
(iv) Dedendum (v) Pitch Circle (vi) Pitch Circle diameter | Remember /
CO2 | 06 |
| B) Classify gears on the basis of various shaft positions. Also name the application of each. Provide neat diagram wherever possible. | Apply / CO2 | 06 |
| C) A pinion having 30 teeth drives a gear having 80 teeth. The profile of the gears is involute with 20° pressure angle, 12 mm module and 10 mm addendum. Find the length of path of contact, arc of contact and the contact ratio. | Evaluating /
CO2 | 06 |

Q. 3 Solve Any Two of the following.

- | | | |
|---|---------------------|-----------|
| A) Explain in short types of gear trains with line diagram. Also state the formula to find velocity ratio in each case. | Understand /
CO2 | 06 |
|---|---------------------|-----------|

- | | | | |
|----|--|---------------------|-----------|
| B) | Describe various types of the torques in an epicyclic gear train and their relationship with neat sketch. | Remember /
CO2 | 06 |
| | | | |
| C) | In an epicyclic gear train, an arm carries two gears A and B having 36 and 45 teeth respectively. If the arm rotates at 150 rpm. in the anticlockwise direction about the centre of the gear A which is fixed, determine the speed of gear B. If the gear A instead of being fixed, makes 300 rpm. in the clockwise direction, what will be the speed of gear B? | Evaluating /
CO2 | 06 |

Q.4 Solve Any Two of the following.

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|----|--|---------------------|-----------|
| A) | Define the following terms relating to governors :
(1) Stability (2) Sensitiveness (3) Isochronism
(4) Hunting (5) Effort of Governor (6) Power of Governor | Remember /
CO3 | 06 |
| | | | |
| B) | Explain the turning moment diagram of a four stroke cycle internal combustion engine. Also explain the role of flywheel in IC Engine. | Understand /
CO4 | 06 |
| | | | |
| C) | The turbine rotor of a ship has a mass of 3500 kg. It has a radius of gyration of 0.45 m and a speed of 3000 rpm clockwise when looking from stern. Determine the gyroscopic couple and its effect upon the ship:
1. When the ship is steering to the left on a curve of 100 m radius at a speed of 36 km/h.
2. When the ship is pitching in a simple harmonic motion, the bow falling with its maximum velocity. The period of pitching is 40 seconds and the total angular displacement between the two extreme positions of pitching is 12 degrees. | Applying /
CO5 | 06 |

Q. 5 Solve Any Two of the following.

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|----|--|---------------------|-----------|
| A) | Discuss briefly with neat sketch the longitudinal, transverse and torsional free vibration. | Remember /
CO6 | 06 |
| | | | |
| B) | Derive an expression for prediction of natural frequency of vibration for single spring mass system. | Understand /
CO6 | 06 |
| | | | |
| C) | Explain and Derive expression for critical or whirling speed of shaft. | Analyzing /
CO6 | 06 |

***** End *****